Universal PLC Firmware Development

Project Overview

Project Name: Universal PLC / IoT Gateway Controller
MCU: STM32F407VGT6
Firmware Environment: STM32CubeIDE + HAL + FreeRTOS
Communication: GSM (EC200U), Ethernet (W5500), RS485, I2C, SPI
Protocols: MQTT, Modbus RTU, Modbus TCP/IP, HTTP (OTA), NTP
PLC Runtime: OpenPLC (Ladder / FBD / STL)

Objectives

- Develop a robust industrial-grade PLC controller
- Modular firmware where **all modules can be enabled/disabled dynamically**
- Support remote configuration, diagnostics, OTA, and cloud connectivity
- Reliable operation with automatic recovery and watchdog supervision
- Seamless integration of I/O, communication, energy monitoring, and PLC logic

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M1 – Project Setup & FreeRTOS Initialization

Tasks

- Setup STM32CubeIDE project for STM32F407VGT6
- Configure system clock, GPIO, UART, SPI, I2C
- Integrate FreeRTOS kernel and scheduler
- Create core tasks:
  - System Supervisor Task
  - I/O Manager Task
  - Communication Manager Task
- Implement UART debug logging
- Task health monitoring and heartbeat mechanism

Deliverables

- Base firmware running FreeRTOS
- UART console output with task status

Acceptance

- Stable scheduler operation
- Task watchdog detects and recovers stalled tasks

M2 – Digital & Analog I/O

Tasks

- Digital Input driver (debounce, edge detect)
- Digital Output driver (OPTO control)
- Timer-based and scheduler-based DO control (single/multiple IO simultaneously)
- ADC (12-bit, DMA) for Analog Inputs:
    - 0–10V / 4–20mA
    - Configurable sampling count (as discussed)
    - Accuracy modes: Weak / Normal / Strong (filtering)
- DAC (MCP4922) for Analog Output:
    - 0–10V / 4–20mA / 0–20mA
    - OPA197 buffer
    - Maximum DAC resolution
- RTD/PT100/PT1000 integration:
    - 3-wire sensing
    - High accuracy temperature measurement

Deliverables

- Verified DI/DO/AI/AO operation
- Calibration parameters stored in non-volatile memory

M3 – GSM (EC200U)+wifi/BT

Tasks

- GSM driver with AT command parser
- Internet connectivity via GSM
- GSM–Ethernet fallback mechanism
- Auto-reconnect when network is restored
- GPS integration:
    - Accuracy within ~10 meters
    - On-demand location response
- GSM loopback mode:
    - GSM provides internet connectivity to Ethernet (W5500) network extension

- Wifi and bluetooth configuration
- web configuration using wifi default id password provide as per teltonika
- wifi network connect MQTT and fallback with gsm and ethernet

Deliverables

- Stable GSM data connection
- GPS data available on request
- Wifi configuration as per teltonika
- All module configuration over the bluetooth

M4 – Ethernet

Tasks

- W5500 Ethernet driver integration
- DHCP by default, configurable Static IP
- Support **multiple IP addresses** on single device:
  - Example: 192.168.1.123, 192.168.10.123, 192.168.34.123
- Modbus TCP/IP (Client + Server simultaneously)
- Ethernet-based I/O expansion
- HMI connectivity via Ethernet
- MQTT over Ethernet
- NTP time synchronization
- TCP/IP based data logger
- All configuration via TCP/IP

Deliverables

- Multi-IP Ethernet stack
- Stable Modbus TCP/IP and MQTT operation

M5 – MQTT

Topics, Payload Format & Tasks

Common Fields (All Topics)

```json
{
  "ts": 1760101200,
  "datetime": "DD/MM/YYYY HH:MM:SS",
  "device_id": "PLC_MAIN_01"
}
```

Topic: `plc/io/data`

```json
{
  "ts": 1760101200,
```

```
  "datetime": "10/10/2025 18:30:00",
  "PLC_IO": {
    "AI": [0.005371, 0.005371, 0.005371, 0.005371],
    "AI_RTD": [0.005371, 0.005371],
    "AO": [0.005371, 0.005371],
    "DIN": [0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0],
    "DOUT": [0,0,0,0,0,0,0,0,0,0,0,0],
    "IO_Status": {
      "AI_Health": "OK",
      "DI_Health": "OK",
      "AO_Health": "OK",
      "DO_Health": "OK"
    }
  }
}
```

Topic: `plc/modbus/tcp`


```
{
  "enabled": true,
  "devices": {
    "PLC_TCP_1": {
      "ip": "192.168.1.100",
      "port": 502,
      "unit_id": 1,
      "ts": 1760101200,
      "txn": "3,40001,10",
      "res": "0,OK",
      "data": [1200,230,15,1,0,32767,32767],
      "datalen": 7
    }
  }
}
```

Topic: `plc/modbus/rtu`


```
{
  "enabled": true,
  "interface": "RS485",
  "baudrate": 9600,
  "parity": "N",
  "devices": {
    "1": {
      "slave_id": 1,
      "ts": 1760101200,
      "device_type": 0,
      "txn": "1,4,1,4",
```

```
      "res": "0,OK",
      "data": [254,230,32767,32767],
      "datalen": 4
    }
  }
}
```

Topic: `plc/energy/data`

```
{
  "meter_id": "EM_01",
  "ts": 1760101200,
  "voltage": {"R":230.1,"Y":229.8,"B":231.0,"avg":230.3},
  "current": {"R":5.21,"Y":4.98,"B":5.10,"avg":5.10},
  "power": {"active_kw":3.45,"reactive_kvar":1.20,"apparent_kva":3.65},
  "power_factor": 0.94,
  "frequency": 49.98,
  "energy_consumption": {
    "kwh_total":12456.78,
    "kvarh_total":3456.12,
    "kva_h_total":14000.55
  }
}
```

Topic: `plc/settings/alarm`

```
{
  "firmware_version": "1.0.1",
  "analog_input": {
    "set_high": 10,
    "over_high": 12,
    "set_low": 2,
    "over_low": 1
  },
  "analog_output": {
    "set_high": 10,
    "over_high": 12,
    "set_low": 2,
    "over_low": 1
  }
}
```

Topic: `plc/system/diagnostics`

```
{
  "cpu_load_percent": 42,
```

```
  "memory_free_kb": 128,
  "uptime_sec": 86400,
  "last_restart_reason": "POWER_ON",
  "faults": []
}
```


MQTT Functional Tasks

- QoS 0/1 support
- Retained messages for configuration topics
- Offline buffering with SD fallback
- Auto-publish after reconnection

M6 – Modbus RTU (RS485)

Tasks

- RS485 driver with direction control
- Multiple slave devices with different baud rates
- Simultaneous polling of multiple slaves
- Master mode:
  - Read/Write registers
- Slave mode:
  - Expose all internal parameters
- SMS-based RS485 control
- HMI connection via RS485:
  - Configuration
  - Live data display
- Error handling:
  - Retry after transaction cycle
  - Slave switching to avoid data loss
  - Last value retention on timeout
  - Alert via MQTT and SMS

M7 – Modbus TCP/IP

Tasks

- Client and Server operation
- Connect devices across same and different IP ranges
- HMI connectivity for configuration and live data
- TCP/IP based I/O expansion
- Data logger access from PC or other devices

M8 – I/O Expansion

Tasks

- I/O expansion via Modbus TCP/IP
- Configure external I/O modules using PLC
- IP configuration support
- Robust retry and timeout handling
- Data retention on timeout
- Alert generation via MQTT and SMS

M9 – I2C

Tasks

- Auto scan for slave addresses
- Auto-connect detected devices
- Dynamic configuration based on slave parameters
- Support additional I2C peripherals

M10 – Energy Meter

Tasks

- Modbus read/write configuration
- SPI driver for ATM90E36A
- Measurement:
    - RMS voltage & current
    - Active/Reactive power
    - Power factor
    - Energy (kWh)
- Calibration support
- MQTT & SD logging integration
- High/Low alert configuration
- SMS alert when GSM is available
- Parameter validation with actual hardware

M11 – OpenPLC Integration

Tasks

- Integrate OpenPLC runtime
- Hardware I/O to PLC variable mapping
- PLC program upload/download (USB / SD / OTA)
- PLC start/stop/reset via HMI or Cloud
- PLC file OTA support

M12 – HMI

Pages

1. Digital & Analog I/O status and control
2. Ethernet & Network configuration
3. GPS status and coordinates
4. Energy meter data
5. Alarm and fault logs

M12A – SMS Configuration & Control (New Milestone)

Scope

All critical configuration, control, and alert mechanisms must be available via **SMS**, independent of Ethernet/GSM data connectivity.

Tasks

- SMS engine over GSM (EC200U)
- Command parser with authentication
- Support **configuration, control, status, and alert SMS**

User Roles

- **Admin Number**
  - Full access
  - Can add/remove user numbers
  - Can change system configuration
  - Can enable/disable modules
- **User Number**
  - Read-only or limited control
  - Can request status and live values

SMS Number Management

- Set primary admin number via:
  - Factory default
  - First valid SMS command
- Support multiple user numbers (configurable limit)
- Store numbers in non-volatile memory

SMS Configuration Commands (Examples)

- Network:
  - Set APN
  - Set Static/DHCP IP
  - Set MQTT broker / port
- I/O:
  - Enable/disable DI/DO/AI/AO channels
  - Set analog scaling
- Modbus:
  - Set RS485 baudrate, parity, slave ID
  - Enable/disable Modbus RTU / TCP
- System:
  - Restart module
  - Restart full system
  - Enable/disable watchdog

SMS Status Commands

- Get system status
- Get I/O live values
- Get energy meter values
- Get network status (GSM/Ethernet)
- Get last fault reason

SMS Alert Messages

Automatically sent to **Admin + configured Users** on: - Analog input over/under limit - Analog output fault - Energy parameter over limit - Modbus communication failure - Network down / restored - System restart / watchdog reset

Security

- Command format validation
- Optional PIN-based authentication
- Ignore unauthorized numbers

Deliverables

- Fully functional SMS control & alert system
- SMS command reference table

Testing & Verification

- Unauthorized number rejection test
- Admin/User privilege validation
- Alert delivery under no-data condition
- GSM network drop & recovery test

M13 – OTA

Tasks

- Dual-bank memory partition:
  - Bootloader
  - Application
- OTA on request (Ethernet/GSM)
- HTTP download with checksum verification
- Bank switching logic
- PLC file OTA support

M14 – SD Card

Tasks

- SD card initialization using FatFS
- Folder structure:
  - `/DATA`
  - `/FAULT`
  - `/CONFIGURATION`
- Daily rotated JSON log files
- Configuration change logging
- Fault logging
- Data buffering when network unavailable
- Automatic publish after connectivity restored
- Configurable buffer publish interval (e.g. 500ms / 1000ms)

M15 – Final Testing & Documentation

Tasks

- 120-hour continuous operation test
- Watchdog and fault recovery validation
- MQTT & JSON payload verification
- Developer Manual
- User Manual (SMS & MQTT commands)
- Architecture flow diagrams
- Task diagrams
- Final handover and walkthrough

Acceptance Criteria Summary

Milestone	Deliverables	Testing & Verification
M1	FreeRTOS base firmware	Task heartbeat, watchdog reset
M2	DI/DO/AI/AO + RTD	Calibration, stress IO toggle
M3	GSM + GPS	Network drop & auto reconnect
M4	Ethernet + Multi-IP	DHCP/static/IP switch tests
M5	MQTT JSON topics	Broker reconnect & payload validation
M6	Modbus RTU	Multi-slave polling, timeout retry
M7	Modbus TCP	Client/server parallel test
M8	IO Expansion	Data integrity & failover
M9	I2C	Auto scan & dynamic config
M10	Energy Meter	Cross-check with reference meter
M11	OpenPLC	Ladder logic IO mapping
M12	HMI	Live data & config update
M13	OTA	Bank switch & rollback
M14	SD Card	Power loss & buffer recovery
M15	Final Test	120-hour burn-in with tamp -10 to +60 degrees

Global Conditions

- All modules runtime enable/disable
- Restart command restarts **all active modules**
- Fault → log → alert (MQTT + SMS)
- Last valid value retained on timeout

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